

Colorado Motor Vehicle Sales Analysis (2008–2015)

Domain: Business Intelligence & Economic Analytics

Objective

This project analyzes historical motor vehicle sales data across Colorado counties to identify trends, seasonal patterns, and economic impacts. It forecasts future sales using advanced time-series models and provides actionable insights for businesses, policymakers, and investors.

Key Deliverables

1. Trend Analysis:
 - Identified a 20% sales decline during the 2008–2009 recession and steady recovery post-2010.
 - Revealed Q4 as the peak sales quarter (holiday demand) and Q1 as the weakest (post-holiday slump).
 2. Regional Insights:
 - Top counties:
Arapahoe (20.1B), ElPaso(20.1B), ElPaso(11.9B).
 - Bottom counties:
Fremont (0.3B), Boulder/Broomfield(0.3B), Boulder/Broomfield(0.7B).
 3. Forecasting:
 - Built a SARIMA model (RMSE: \$0.15B, R^2 : 0.72) to predict quarterly sales.
 - Forecasted 2016 sales growth with 95% confidence intervals.
 4. Business Recommendations:
 - Align inventory with seasonal peaks (Q2/Q4).
 - Target high-growth counties (e.g., Denver) and promote EVs in urban areas.
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Methodology

1. Data Preparation:
 - Imported libraries, aggregated quarterly sales.
2. Exploratory Analysis:
 - Visualized trends using line charts, heatmaps, and YoY growth analysis.
3. Modeling:
 - Applied seasonal decomposition to isolate trends and cyclical patterns.
 - Trained SARIMA model with cross-validation for robust forecasting.
4. Validation:
 - Residual analysis confirmed no autocorrelation and normally distributed errors.

Tools & Technologies

- Python Libraries: Pandas, Matplotlib, Seaborn, Statsmodels.
 - Modeling: SARIMA (Seasonal ARIMA).
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Impact

- Businesses: Optimize inventory, target high-growth regions, and align marketing with seasonal trends.
 - Policymakers: Design EV incentives and infrastructure plans for underperforming counties.
 - Investors: Identify growth opportunities using predictive insights.
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Conclusion

By transforming raw sales data into strategic insights, this project bridges data science with real-world automotive industry challenges. Its scalable framework supports future enhancements, such as integrating macroeconomic indicators (e.g., fuel prices) for richer analysis.